

Qn	Suggested Solution
1(i)	$a_n = \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} + \dots$ $< \frac{1}{n+1} + \frac{1}{(n+1)^2} + \frac{1}{(n+1)^3} + \dots$ $= \frac{1}{n+1} \left(\frac{1}{1 - \frac{1}{n+1}} \right)$ $= \frac{1}{n+1} \left(\frac{n+1}{n+1-1} \right)$ $= \frac{1}{n}$
1(ii)	LHS = e $= 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$ $= 2 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$ $= \lfloor e \rfloor + a_1$ = RHS LHS = 2e $= 2 \left(1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \right)$ $= 2 + 2 + 1 + \frac{1}{3} + \frac{1}{(3)(4)} + \frac{1}{(3)(4)(5)} + \dots$ $= \lfloor 2e \rfloor + a_2$ = RHS
1(iii)	$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots + \frac{1}{n!} + \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \dots$ $n!e = n! + n! + \frac{n!}{2!} + \frac{n!}{3!} + \frac{n!}{4!} + \dots + \frac{n!}{n!} + \frac{n!}{(n+1)!} + \frac{n!}{(n+2)!} + \dots$ $= n! + n! + \frac{n!}{2!} + \frac{n!}{3!} + \frac{n!}{4!} + \dots + \frac{n!}{n!} + a_n$ Since $n! + n! + \frac{n!}{2!} + \frac{n!}{3!} + \frac{n!}{4!} + \dots + \frac{n!}{n!}$ gives an integer and that $0 < a_n < \frac{1}{n} \leq 1 \text{ for each positive integer } n, \text{ therefore}$ $n! + n! + \frac{n!}{2!} + \frac{n!}{3!} + \frac{n!}{4!} + \dots + \frac{n!}{n!} = \lfloor n!e \rfloor.$ Hence we have $a_n = n!e - \lfloor n!e \rfloor.$

Alternatively,

Let P_n denote “ $a_n = n!e - \lfloor n!e \rfloor$ ” for $n = 1, 2, 3, \dots$

When $n = 1$,

$$\text{LHS of } P_1 = a_1 = \frac{1}{2} + \frac{1}{(2)(3)} + \frac{1}{(2)(3)(4)} + \dots$$

$$\text{RHS of } P_1 = e - \lfloor e \rfloor = \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$$

Therefore P_1 is true.

Suppose P_k is true for some $k = 1, 2, 3, \dots$

LHS of $P_k = a_{k+1}$

$$= \frac{1}{k+2} + \frac{1}{(k+2)(k+3)} + \frac{1}{(k+2)(k+3)(k+4)} + \dots$$

$$= (k+1) \left[\frac{1}{(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} + \dots \right]$$

$$= (k+1) \left(a_k - \frac{1}{k+1} \right) = (k+1) (k!e - \lfloor k!e \rfloor) - 1$$

$$= (k+1)!e - (k+1)\lfloor k!e \rfloor - 1 = (k+1)!e - \lfloor (k+1)!e \rfloor$$

$$= \text{RHS of } P_{k+1}$$

The final equality holds because

$$(k+1)!e = (k+1)! \left(2 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{(k+1)!} + \frac{1}{(k+2)!} + \dots \right)$$

$$= (k+1)! \left(2 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{(k+1)!} \right) + a_2$$

$$\because 0 < a_2 < \frac{1}{2} < 1 \text{ and } (k+1)! \left(2 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{(k+1)!} \right) \in \mathbb{Z}$$

$$\therefore \lfloor (k+1)!e \rfloor = (k+1)! \left(2 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{(k+1)!} \right)$$

$$(k+1)\lfloor k!e \rfloor + 1 = (k+1)(k!) \left(2 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} \right) + 1$$

$$= (k+1)! \left(2 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} \right) + 1$$

$$= (k+1)! \left(2 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} + \frac{1}{(k+1)!} \right)$$

Since P_1 is true, and P_k is true implies that P_{k+1} is true, then by mathematical induction, P_n is true for $n = 1, 2, 3, \dots$

1(iv)	Suppose e is a rational number. Then we can express $e = \frac{p}{q}$, where p and q are integers. This implies that $eq = p$ which is an integer, which in turn means that $e(q!)$ is also an integer. Therefore $\lfloor e(q!) \rfloor = e(q!) \Rightarrow e(q!) - \lfloor e(q!) \rfloor = 0$, which is a contradiction (based on part ii which is proven that $e(q!) - \lfloor e(q!) \rfloor > 0$). e is an irrational number.
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Qn	Suggested Solution
2(i)	$\theta = \tan^{-1} \frac{b}{x} - \tan^{-1} \frac{a}{x}$ $\frac{d\theta}{dx} = \left(\frac{-b}{x^2} \right) \left(\frac{1}{1 + \left(\frac{b}{x} \right)^2} \right) - \left(\frac{-a}{x^2} \right) \left(\frac{1}{1 + \left(\frac{a}{x} \right)^2} \right)$ $= \frac{a}{x^2 + a^2} - \frac{b}{x^2 + b^2}$ When $\frac{d\theta}{dx} = 0$, $\frac{a}{x^2 + a^2} - \frac{b}{x^2 + b^2} = 0$ $\frac{a}{x^2 + a^2} = \frac{b}{x^2 + b^2}$ $ax^2 + ab^2 = bx^2 + a^2b$ $(a - b)x^2 = (a - b)ab$ $x^2 = ab$ $x = \sqrt{ab} \quad \text{or} \quad x = -\sqrt{ab} \quad (\text{rej. } \because x > 0)$ $\frac{d^2\theta}{dx^2} = \frac{2bx}{(x^2 + b^2)^2} - \frac{2ax}{(x^2 + a^2)^2}$ When $x = \sqrt{ab}$, $\frac{d^2\theta}{dx^2} = \frac{2b\sqrt{ab}}{(ab + b^2)^2} - \frac{2a\sqrt{ab}}{(ab + a^2)^2}$ $= \frac{2b\sqrt{ab}}{b^2(a + b)^2} - \frac{2a\sqrt{ab}}{a^2(a + b)^2}$ $= \frac{2a^2b\sqrt{ab} - 2ab^2\sqrt{ab}}{a^2b^2(a + b)^2}$ $= \frac{2ab\sqrt{ab}(a - b)}{a^2b^2(a + b)^2} < 0 \quad (\because a < b)$ Therefore, a local maximum at $x = \sqrt{ab}$. Since there are no other stationary points, θ achieves a global maximum at $x = \sqrt{ab}$.
2(ii)	Since $\tan \theta$ is strictly increasing when θ is acute, the maximum value of $\tan \theta$ also occurs at $x = \sqrt{ab}$. Let $\tan^{-1} \frac{a}{x} = \alpha$ and $\tan^{-1} \frac{b}{x} = \beta$. Thus, $\tan \alpha = \frac{a}{x}$ and $\tan \beta = \frac{b}{x}$

	$\tan \theta = \tan(\beta - \alpha)$ $= \frac{\tan \beta - \tan \alpha}{1 + \tan \beta \tan \alpha}$ $= \frac{\frac{b}{x} - \frac{a}{x}}{1 + \frac{ab}{x^2}}$ $= \frac{(b-a)x}{x^2 + ab}$ When $x = \sqrt{ab}$, $\tan \theta = \frac{(b-a)\sqrt{ab}}{ab + ab}$ $= \frac{(b-a)\sqrt{ab}}{2ab} \quad \text{or} \quad \frac{b-a}{2\sqrt{ab}}$
2(iii)	Let the centre of the circle passing through the points P, Q and S be O. O must lie on the perpendicular bisector of PQ. Thus, the O lies $\frac{a+b}{2}$ units above SR. Observe that OP and OQ are radii of the circle. By Pythagoras' Theorem, $OP^2 = (\sqrt{ab})^2 + \left(\frac{b-a}{2}\right)^2$ $= ab + \frac{a^2 + b^2 - 2ab}{4}$ $= \frac{a^2 + b^2 + 2ab}{4}$ $= \left(\frac{a+b}{2}\right)^2$ $OP = \frac{a+b}{2}$ Since the radius of the circle is $\frac{a+b}{2}$, O lies vertically above S. Thus, $OS \perp SR$ and therefore SR is tangent to the circle passing through the points P, Q and S.

Qn	Suggested Solution
3(i)	Let f and g be continuous and convex functions on $\mathbb{R}$. Thus for all $a, b \in \mathbb{R}$ and $\lambda \in [0, 1]$, $f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b) \text{ and}$ $g(\lambda a + (1 - \lambda)b) \leq \lambda g(a) + (1 - \lambda)g(b).$ Summing both inequalities, $f(\lambda a + (1 - \lambda)b) + g(\lambda a + (1 - \lambda)b)$ $\leq \lambda f(a) + (1 - \lambda)f(b) + \lambda g(a) + (1 - \lambda)g(b)$ $(f + g)(\lambda a + (1 - \lambda)b)$ $\leq \lambda [f(a) + g(a)] + (1 - \lambda)[f(b) + g(b)]$ $(f + g)(\lambda a + (1 - \lambda)b) \leq \lambda (f + g)(a) + (1 - \lambda)(f + g)(b)$ Therefore, $f + g$ is a convex function.
3(ii)	Let $f(x) = kx^2$ where $k > 0$, $\lambda \in [0, 1]$ and $p, q \in \mathbb{R}$, $f(\lambda a + (1 - \lambda)b)$ $= k[\lambda a + (1 - \lambda)b]^2$ $= k[\lambda^2 a^2 + (1 - \lambda)^2 b^2 + 2\lambda(1 - \lambda)ab] \text{ ---- (1)}$ On the other hand, $\lambda f(a) + (1 - \lambda)f(b)$ $= \lambda ka^2 + (1 - \lambda)kb^2$ $= k[\lambda a^2 + (1 - \lambda)b^2] \text{ ---- (2)}$ Taking (2) – (1): $\lambda f(a) + (1 - \lambda)f(b) - f(\lambda a + (1 - \lambda)b)$ $= k[\lambda a^2 + (1 - \lambda)b^2] - k[\lambda^2 a^2 + (1 - \lambda)^2 b^2 + 2\lambda(1 - \lambda)ab]$ $= k[\lambda a^2(1 - \lambda) + (1 - \lambda)b^2(1 - (1 - \lambda)) - 2\lambda(1 - \lambda)ab]$ $= k\lambda(1 - \lambda)(a^2 + b^2 - 2ab)$ $= k\lambda(1 - \lambda)(a - b)^2$ $\geq 0 \quad (\because a > 0, \lambda \geq 0, 1 - \lambda \geq 0 \text{ and } (a + b)^2 \geq 0)$ Therefore, $f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b)$
3(iii) (a)	Let $f(x) = mx + c$, $\lambda \in [0, 1]$ and $a, b \in \mathbb{R}$ $f(\lambda a + (1 - \lambda)b) = m(\lambda a + (1 - \lambda)b) + c$ $= \lambda ma + (1 - \lambda)mb + c$

	$\lambda f(a) + (1-\lambda)f(b)$ $= \lambda(ma+c) + (1-\lambda)(mb+c)$ $= \lambda ma + (1-\lambda)mb + c$ $= f(\lambda a + (1-\lambda)b)$ $\geq f(\lambda a + (1-\lambda)b)$ Therefore, linear functions are convex.
3(iii) (b)	Let $f(x) = x$ and $g(x) = x^2$ which are both convex functions. Thus we have $(fg)(x) = x^3$ Then, $f(0.5(-3) + 0.5(-1)) = f(-2)$ $= -8$ On the other hand, $0.5f(-3) + 0.5f(-1) = 0.5(-27) + 0.5(-1)$ $= -14$ < -8 $= f(0.5(-3) + 0.5(-1))$ Therefore, it is not true that “the product of two convex functions is convex”.
3(iv)	Let f be a function strictly convex on $\mathbb{R}$ and $\lambda \in [0,1]$. Suppose that f achieves its minimum value at $x = a$ and $x = b$ where $a \neq b$. Then, $f(\lambda a + (1-\lambda)b) < \lambda f(a) + (1-\lambda)f(b)$ $= \lambda f(a) + (1-\lambda)f(a)$ $= f(a)$ which is a contradiction since $f(a)$ is the minimum value of the function.

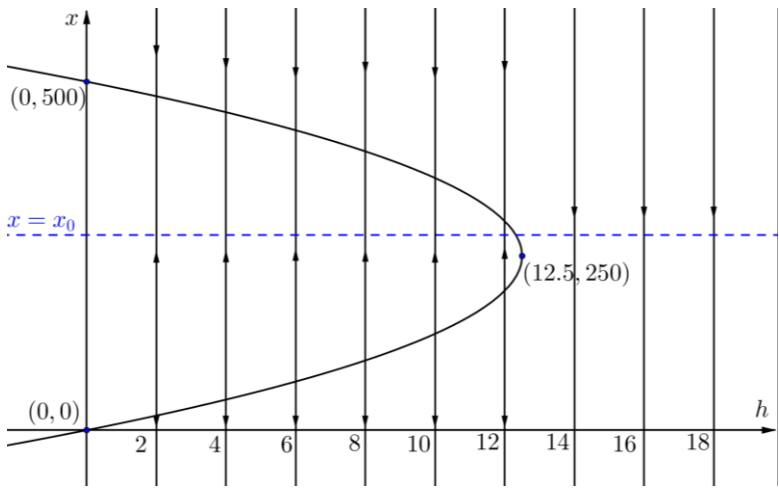
Qn	Suggested Solution
4(a) (i)	$(1, 0, 0), (-1, 0, 0), (0, 1, 0), (0, -1, 0), (0, 0, 1), (0, 0, -1)$ $T_1 = 6$
4(a) (ii)	Consider the following cases: <u>Case 1: All the 3 numbers are not zero</u> (Distributing n identical objects into 3 distinct boxes with no empty box) The number of such triples are $2^3 \binom{n-1}{2} = 4(n-1)(n-2)$ (each number can be either positive or negative) <u>Case 2: Exactly one of the 3 numbers is zero</u> The number of such triples are $\binom{3}{1} 2^2 \binom{n-1}{1} = 12(n-1)$ <u>Case 3: Exactly two of the 3 numbers are zero</u> The number of such triples are $\binom{3}{2} (2)(1) = 6$ Thus, $T_n = 4(n-1)(n-2) + 12(n-1) + 6 = 4n^2 + 2$ by AP.
4(b) (i)	<u>Step 1: Selecting two numbers to be placed in the same subset</u> Number of ways = $\binom{n}{2} = \frac{n(n-1)}{2}$ <u>Step 2: Placing the remaining $(n-2)$ numbers in the remaining subsets</u> Number of ways = 1 Thus, $S(n, n-1) = \frac{n(n-1)}{2}$
4(b) (ii)	Considering the following cases: <u>Case 1: Exactly 3 of the numbers are placed in one subset</u> Number of ways = $\binom{n}{3} = \frac{n(n-1)(n-2)}{6}$ <u>Case 2: Exactly 2 of the numbers are placed in one subset and another 2 of the numbers placed are in another subset</u> Number of ways = $\binom{n}{4} \binom{3}{1} = \frac{n(n-1)(n-2)(n-3)}{8}$ By AP, $S(n, n-2) = \frac{n(n-1)(n-2)}{6} + \frac{n(n-1)(n-2)(n-3)}{8}$ $= \frac{4n(n-1)(n-2) + n(n-1)(n-2)(3n-9)}{24}$ $= \frac{n(n-1)(n-2)(3n-5)}{24}$
Qn	Suggested Solution
5(a)	$12^{50} = 2^{100} \times 3^{50}, 45^{20} = 3^{40} \times 5^{20}, 50^{15} = 2^{15} \times 5^{30}.$

	Let A denote the set of positive divisors of 12^{50}, $A = 101 \times 51 = 5151$ Let B denote the set of positive divisors of 45^{20}, $B = 41 \times 21 = 861$ Let C denote the set of positive divisors of 50^{15}, $C = 16 \times 31 = 496$ $A \cap B$ denote the set of common divisors of 12^{50} and 45^{20}, $A \cap B = 41$ $B \cap C$ denote the set of common divisors of 45^{20} and 50^{15}, $B \cap C = 21$ $C \cap A$ denote the set of common divisors of 50^{15} and 12^{50}, $C \cap A = 16$ $A \cap B \cap C$ denote the set of common divisors of all the 3 numbers, $A \cap B \cap C = 1$. By PIE, $A \cup B \cup C$ $= A + B + C - A \cap B - B \cap C - C \cap A + A \cap B \cap C$ $= 5151 + 861 + 496 - 41 - 21 - 16 + 1$ $= 6431$
5(b)(i)	$a_1 = 1$, $a_2 = 1$ and $a_3 = 2$
5(b)(ii)	Consider the following cases for filling a $3 \times n$ checkerboard <u>Case 1: The domino at the top left corner is placed vertically</u> Number of ways to fill the remaining $3 \times (n-1)$ checkerboard is a_{n-1}. <u>Case 2: The domino at the top left corner is placed horizontally</u> Two more dominos have to be placed directly below it. Number of ways to fill the remaining $3 \times (n-3)$ checkerboard is a_{n-3}. By AP, $a_n = a_{n-1} + a_{n-3}$, $n \geq 4$ $a_4 = a_3 + a_1 = 3$, $a_5 = a_4 + a_2 = 4$, $a_6 = a_5 + a_3 = 6$, $a_7 = a_6 + a_4 = 9$, $a_8 = a_7 + a_5 = 13$, $a_9 = a_8 + a_6 = 19$
5(b)(iii)	Draw a line at the centre of the $3 \times 2n$ checkerboard, dividing it into two $3 \times n$ checkerboards. Consider the following cases for filling this checkerboard: <u>Case 1: no domino crosses the middle line</u> The number of ways to fill each of the $3 \times n$ checkerboards is a_n, so the number of ways is a_n^2.

	Case 2: <u>some dominos cross the middle line</u> Let the number of ways be b. In a scenario that we place 3 dominos horizontally such that each domino has 1 square is on the left and 2 squares on the right of the middle line, the number of ways is $(a_{n-1})(1)(a_{n-2}) \geq 1.$ Thus $b \geq 1$. By AP, $a_{2n} = a_n^2 + b > a_n^2$
5 Last Part	Each domino will cut exactly two lines (either horizontal or vertical) The number of dominos needed to fill up the checkerboard is $\frac{9 \times 15}{3 \times 1} = 45$, these dominos will cut the lines exactly 90 times. The total number of lines (both horizontal and vertical) is $8 + 14 = 22$. Since $90 = 22 \times 4 + 2$ (or $\left\lceil \frac{90}{22} \right\rceil = 5$), by Pigeonhole Principle, there exists at least one line which will be cut by 5 dominos.

Qn	Suggested Solution
6(i)	$\frac{ds}{d\theta} - s \tan \theta = as^3 \sec \theta$ $u = s^{-2} \Rightarrow \frac{du}{d\theta} = -2s^{-3} \frac{ds}{d\theta}$ $\therefore -\frac{1}{2}s^3 \frac{du}{d\theta} - s \tan \theta = as^3 \sec \theta$ $\Rightarrow -\frac{1}{2}s^2 \frac{du}{d\theta} - \tan \theta = as^2 \sec \theta$ $\Rightarrow \frac{du}{d\theta} + 2u \tan \theta = -2a \sec \theta$ Integrating factor $= e^{\int 2 \tan \theta d\theta}$ $= e^{2 \ln(\sec \theta)}$ $= \sec^2 \theta$ Hence, $\frac{d}{d\theta}(u \sec^2 \theta) = -2a \sec^3 \theta$ $u \sec^2 \theta = -2a \int \sec^3 \theta d\theta$ $\int \sec^3 \theta d\theta = \int \sec \theta \sec^2 \theta d\theta$ $= \sec \theta \tan \theta - \int \tan \theta (\sec \theta \tan \theta) d\theta$ $= \sec \theta \tan \theta - \int \sec \theta \tan^2 \theta d\theta$ $= \sec \theta \tan \theta - \int (\sec^3 \theta - \sec \theta) d\theta$ $2 \int \sec^3 \theta d\theta = \sec \theta \tan \theta + \int \sec \theta d\theta$ $= \sec \theta \tan \theta + \ln(\sec \theta + \tan \theta) + C'$ $\int \sec^3 \theta d\theta = \frac{\sec \theta \tan \theta + \ln(\sec \theta + \tan \theta) + C'}{2}$ $\therefore u \sec^2 \theta = -a [\sec \theta \tan \theta + \ln(\sec \theta + \tan \theta) + C']$ $s^{-2} \sec^2 \theta = -a [\sec \theta \tan \theta + \ln(\sec \theta + \tan \theta) + C']$ $\Rightarrow s^2 = \frac{\sec^2 \theta}{-a [\sec \theta \tan \theta + \ln(\sec \theta + \tan \theta) + C']}$ $= \frac{\sec^2 \theta}{a [m - \sec \theta \tan \theta - \ln(\sec \theta + \tan \theta)]}, m = -C'$ $\therefore s = \frac{\sec \theta}{\sqrt{a [m - \sec \theta \tan \theta - \ln(\sec \theta + \tan \theta)]}} \quad (\text{Shown})$ $\left(\because \sec \theta > 0 \text{ and } \tan \theta > 0 \text{ for } 0 < \theta < \frac{\pi}{2} \right)$

6(ii)	For small θ such that terms in θ^2 and higher powers can be ignored, $\sec \theta \approx \left(1 - \frac{\theta^2}{2}\right)^{-1} \approx 1 \text{ and } \tan \theta \approx \theta.$ $s = \frac{\sec \theta}{\sqrt{a[m - \sec \theta \tan \theta - \ln(\sec \theta + \tan \theta)]}}$ $\approx \frac{1}{\sqrt{a}}(1)[m - \theta - \ln(1 + \theta)]^{-\frac{1}{2}}$ $\approx \frac{1}{\sqrt{a}}(m - \theta - \theta)^{-\frac{1}{2}}$ $= \frac{1}{\sqrt{am}}\left(1 - \frac{2}{m}\theta\right)^{-\frac{1}{2}}$ $\approx \frac{1}{\sqrt{am}}\left(1 + \frac{1}{m}\theta\right) = \frac{1}{m\sqrt{am}}\theta + \frac{1}{\sqrt{am}} \text{ (Shown)}$
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Qn	Suggested Solution
7(a)	 The equilibrium point $y = 2$ is an unstable equilibrium point OR a source. The equilibrium point $y = -2$ is a stable equilibrium point OR a sink.
7(b) (i)	Model with harvesting is $\frac{dx}{dt} = 0.1x - 0.0002x^2 - h$. At the bifurcation value, $(0.1)^2 - 4(-0.0002)(-h) = 0$ $h = 12.5$ 
7(b) (ii)	$\frac{dx}{dt} = 0.1x - 0.0002x^2 - 12$ The equilibrium points occur when $0.1x - 0.0002x^2 - 12 = 0$ $x = 200 \text{ or } x = 300$ From the bifurcation diagram, for the population size to steadily increase towards a fixed number, $200 < x_0 < 300$

7(b)
(iii)

$$\frac{dx}{dt} = 0.1x - 0.0002x^2 - 12$$

$$\frac{dx}{dt} = -0.0002(x-200)(x-300)$$

$$\frac{1}{(x-200)(x-300)} \frac{dx}{dt} = -0.0002$$

$$\frac{100}{(x-200)(x-300)} \frac{dx}{dt} = -0.02$$

$$\int \frac{1}{x-300} - \frac{1}{x-200} dx = \int -0.02 dt$$

$$\ln \left| \frac{x-300}{x-200} \right| = -0.02t + c$$

$$\frac{x-300}{x-200} = \pm e^{-0.02t+c} = Ae^{-0.02t}, A = \pm e^c$$

When $t = 0$, $x = 250$. Hence, $\frac{250-300}{250-200} = A \Rightarrow A = -1$.

$$\frac{x-300}{x-200} = -e^{-0.02t}$$

$$x-300 = -e^{-0.02t}(x-200)$$

$$x(1+e^{-0.02t}) = 300 + 200e^{-0.02t}$$

$$x = \frac{300 + 200e^{-0.02t}}{1+e^{-0.02t}} = 200 + \frac{100}{1+e^{-0.02t}}$$

Qn	Suggested Solution
8(i)	Net force on the body = $0.5 \ddot{x}$ Damping force on the body = $-c\dot{x}$ for some positive constant c Restoring force = $-kx$, where k is the spring constant Thus, $0.5 \ddot{x} = -c\dot{x} - kx$ $\Rightarrow \ddot{x} + 2c\dot{x} + 2kx = 0, \text{ with } a = 2c \text{ and } b = 2k.$
8(ii)	$\ddot{x} + 2c\dot{x} + 4x = 0$, where c is the damping constant. For the motion to be over-damped, $(2c)^2 - 4(4) > 0$ $\Rightarrow c^2 > 4$ $\Rightarrow c > 2 \text{ (since } c \text{ is positive)}$
8(iii)	$\ddot{x} + 3\dot{x} + 8.5x = 0$ The characteristic equation is $m^2 + 3m + 8.5 = 0$. $\Rightarrow m = \frac{-3 \pm \sqrt{3^2 - 4(8.5)}}{2}$ $\Rightarrow m = \frac{-3 \pm \sqrt{-25}}{2} = \frac{-3 \pm 5i}{2}$ $\therefore x = e^{-\frac{3}{2}t} \left[A \cos\left(\frac{5}{2}t\right) + B \sin\left(\frac{5}{2}t\right) \right]$ When $t = 0, x = 0.2 \Rightarrow A = 0.2$ $\dot{x} = -\frac{3}{2}e^{-\frac{3}{2}t} \left[0.2 \cos\left(\frac{5}{2}t\right) + B \sin\left(\frac{5}{2}t\right) \right]$ $+ e^{-\frac{3}{2}t} \left[-0.5 \sin\left(\frac{5}{2}t\right) + \frac{5B}{2} \cos\left(\frac{5}{2}t\right) \right]$ When $t = 0, \dot{x} = 0$ $\Rightarrow 0 = -\frac{3}{2}(0.2) + \frac{5B}{2}$ $\Rightarrow B = \frac{3}{25}$ $\therefore x = e^{-\frac{3}{2}t} \left[\frac{1}{5} \cos\left(\frac{5}{2}t\right) + \frac{3}{25} \sin\left(\frac{5}{2}t\right) \right]$

